ReadMe Document

This ReadMe document will help you with a quick navigation through the contents of the Pre-Reading material, Preview videos, Individual Labs and Mini-Hackathon/Hackathon for the upcoming session

Outcomes

After completion of this week's sessions, & activity participants will be able to :

1. Understand the underlying mathematics behind PCA (Unsupervised learning) for dimensionality reduction and its applications
2. Understand the concept of LDA (Supervised learning) for dimensionality reduction
3. Understand the concept of generalization in ML, problems of overfitting and controlling them
4. Understand & implement evaluation metrics used in classification problems like Precision, Recall & F1 score

Lecture Topics

1. Intro to PCA
2. PCA and Eigen Faces
3. Overfitting and Generalization
4. Performance Metrics

Preview Videos

The Preview Videos Link for the respective sessions are available a week in advance (i.e. every Monday), to help you understand the concepts that will be covered in the upcoming lectures. There could be one or more videos depending on the lecture details. The video/videos should be viewed before attending the lecture for better understanding.

There are 5 preview videos for this session and they should be viewed as given below:

1. Principal Component Analysis: This is a 12-minutes 2 sec video. This video helps you to understand covariance, eigenvalues and eigenvectors. You will also understand how high dimensional data is reduced using PCA.

2. Eigen Faces : This is a 14 minute video. This video helps you to understand about extracting the Eigenvectors using PCA from the given set of images called Eigenfaces. Eigenfaces are a set of features that captures the variance along different orthogonal directions in the image space.

3. Overfitting : This is a 4-minute video. In this video, you will understand how overfitting happens when a model learns the details of the training data to the extent that it negatively impacts the performance of the model on new data.

4. Performance Metrics: This consists of two videos a) Accuracy: This is a 10-minute video. From this video you will learn about what accuracy is. It is a general measure for determining the model performance, but there are more sophisticated metrics. b) Precision and Recall: This is an 8-minute video. This video helps you to understand how True Positive (TP), False Positive (FP), True Negative (TN), False Negative (FN) can be used to determine the metrics Precision and Recall.

Pre Reading Material

The pre-reading material is given on (1) Eigenfaces (2) Overfitting, . You should read the material before you attend the lecture. This pre-reading will provide the gist of the lecture and facilitate your understanding during the lecture.

Reference Material

The Reference Material is given on (1) Principal Component Analysis (2) Eigenfaces (3) Overfitting.. You should read the material after attending the lecture. This reference material will provide further depth into the topic.

Individual Labs

1. Intro to PCA
2. PCA and Eigen Faces
3. Overfitting and Generalization
4. Performance Metrics